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Probing initial state clustering through photon anisotropic flow in 7A TeV ^{16}O+^{16}O collisions at the LHC

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Abstract The presence of α clustered structures in light nuclei can enhance the initial spatial anisotropies in relativistic nuclear collisions relative to those arising from nuclei with uniform density distributions. Thus, observables that are strongly sensitive to the initial geometry can be a more efficient probe of the clustered structures than observables dominated by final state dynamics. We investigate the collisions of α clustered oxygen nuclei at $\sqrt{s_{NN}} = 7$A TeV at the LHC using the GLISSANDO initial state model along with the MUSIC event-by-event hydrodynamical framework. The tetrahedral α clustered structure of ^{16}O leads to significantly larger initial triangular eccentricity ϵ_3 than collisions with uniform density distributions especially in the most central events. The spatial eccentricity ϵ_2 is found to be relatively less sensitive to the initial state clustered structure. The production of thermal photons is estimated to be only marginally influenced by clustering for both central as well as peripheral collisions. In contrast, the photon triangular flow coefficient $v_3(p_T)$ is strongly affected by initial state clustering resulting in substantially larger values in both central and peripheral collisions. An experimental determination of photon anisotropic flow together with the ratios of flow coefficients in ^{16}O+^{16}O collisions therefore expected to provide valuable insight into the possible clustered structure in light nuclei and also to constrain parameters in theoretical modeling.

1 Introduction

Alpha clustering is a well known phenomenon studied in nuclear structure research where the formation of α substructures is known to enhance stability in light nuclei [1–4]. Recent studies suggest that the presence of α clustered structures in light nuclei, such as in ^{12}C and ^{16}O can be effectively investigated at relativistic energies by measuring the anisotropic flow parameters [5–19].

The clustered configurations of light nuclei lead to pronounced spatial anisotropies in the overlap region when they collide with a heavy nuclei at relativistic energies. In addition, collisions involving two clustered nuclei are also expected to significantly influence the initial energy density distribution on the transverse plane as well as the spatial anisotropy of the overlapping zone. Since the emission of electromagnetic radiation is expected to be sensitive to the initial state, the production and anisotropic flow of direct photons can be a promising probe for studying the α clustered structures [20–36]. Additionally, the initial geometry driven effects are expected to be more prominent in the most central collisions compared to the peripheral ones [35].

It has been demonstrated that collisions of triangular α clustered ^{12}C nuclei with Au at the top RHIC energy generate distinct initial geometries in the transverse plane depending on the orientations of the incoming clustered nuclei in central collisions [35]. The initial anisotropic energy density distribution in these collisions lead to a significantly enhanced anisotropic flow ($v_n(p_T)$) of thermal photons which shows strong dependence on the orientation angle. In contrast, the α clustered structure was found to have only a marginal effect

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on the thermal photon spectra obtained at different orientations of C+Au collisions.

A more realistic calculation shows that the orientation averaged anisotropic flow of thermal photons in clustered ^{12}C+Au collisions at 200A GeV exhibits significant qualitative differences between the v_2 and v_3 parameters. The initial triangular eccentricity ϵ_3 in the case of clustered ^{12}C is estimated to be substantially larger compared to that in unclustered ^{12}C+Au collisions [36]. This enhancement translates into a significantly larger triangular flow parameter $v_3(p_T)$ of photons for the clustered case. In contrast, the elliptic flow parameter is observed to be much smaller than the triangular flow in the clustered collisions.

The recent experimental run from ^{16}O+^{16}O collisions at the LHC is expected to provide valuable insights into the formation of a hot and dense medium in small systems as well as into the role of clustered structures in light nuclei [37–41]. These results will also help determine whether investigating clustered structures requires collisions of light nuclei with heavy ions or if collisions between two light clustered nuclei are sufficient to reveal clear signatures of the clustered configuration in the initial state.

Direct photons produced in relativistic nuclear collisions comprise contributions from different stages of the fireball evolution [21,31,42]. The low p_T region of the spectrum is dominated by thermal radiation, which directly contributes to the anisotropic flow of photons. In contrast, prompt photons produced in initial hard scatterings dominate the higher p_T region. Since prompt photons are emitted before the development of collective behavior in the medium, they do not directly contribute to photon anisotropic flow.

Recent studies have shown that photons produced during the pre-equilibrium stage, before the medium thermalizes, can contribute significantly to the direct photon spectrum, particularly at higher transverse momentum [43–47]. It has also been demonstrated in some studies that even with an extended pre-equilibrium phase, the corresponding contribution to photon anisotropic flow remains modest [44,47].

In the present work, we study the evolution of α clustered ^{16}O+^{16}O collisions at 7A TeV at LHC using MUSIC hydrodynamical model framework [49]. The GLISSANDO initial conditions is used to construct the initial state for clustered and unclustered collisions of oxygen nuclei [50]. We use the state of the art photon rates to estimate the thermal photon production and anisotropic flow parameters for two different centrality bins at the LHC energy.

2 Formalism

We use GLISSANDO (Glauber Initial State Simulation AND mOre), a Monte Carlo-based framework to simulate the initial state in α clustered O+O collisions at $\sqrt{s_{NN}} = 7$A TeV at the LHC [50–52]. The framework provides the spatial distributions of participant nucleons in different centrality bins for both clustered and unclustered configurations of ^{16}O.

For the clustered configuration, the ^{16}O nucleus is modeled as a tetrahedral arrangement of four α clusters, with each cluster positioned at a vertex of the tetrahedron. The side length l of the tetrahedron is taken as 3.2 fm and the radius of each cluster r_α is 1.1 fm [53,54]. The distribution of nucleons in each cluster is expressed through a Gaussian

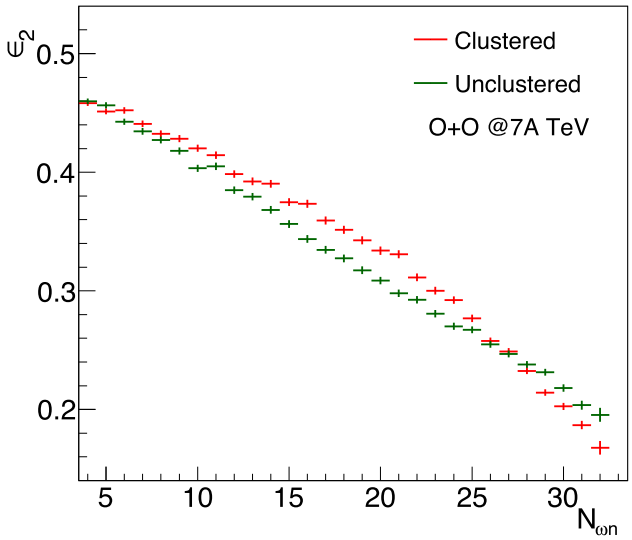


Fig. 1 Initial spatial eccentricity ϵ_2 for clustered and unclustered O+O collisions at 7A TeV at LHC as a function of number of wounded nucleons $N_{\rm WN}$

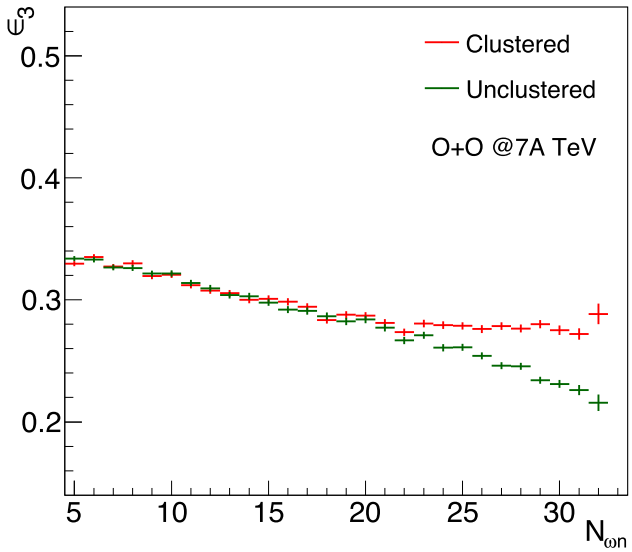


Fig. 2 Initial triangular eccentricity ϵ_3 for clustered and unclustered O+O collisions at 7A TeV at LHC as a function of number of wounded nucleons $N_{\rm WN}$

function as follows:

$$f_i(\vec{r}) = A \exp\left(-\frac{3}{2}(\vec{r} - \vec{c}_i)^2/r_\alpha^2\right). \quad (1)$$

Here, $\vec{c}_i$ denotes the position of the center of ith cluster in oxygen. The distance between the centers of any nucleon-nucleon pair is not to be less than 0.9 fm due to the short-range repulsion [6]. The radial distribution of nucleon centers is used as a constraint when constructing the clustered configurations. For the unclustered case, an isotropic nuclear distribution whose root mean square (rms) radius is set equal to that of the clustered oxygen (∼2.4 fm).

A two-component Monte Carlo Glauber model is used to determine the initial entropy density on the transverse plane [36]:

$$s(x,y) = K \sum_{i,j=1}^{N_{\rm part},N_{\rm coll}} [(1-\alpha) n_{\rm part}(x_i,y_i)\, F_i(x,y) + \alpha\, n_{\rm coll}(x_j,y_j)\, F_j(x,y)]. \quad (2)$$

In the above equation, $n_{\rm part}(x,y)$ and $n_{\rm coll}(x,y)$ denote the number of participants and the number of collisions at the position (x,y) respectively. A participant is given a weight of $(1-\alpha)$ and a binary collision is given a weight of α, where α is taken as 0.12. The function $F_i(x,y)$ ($F_j(x,y)$) is a normalized Gaussian distribution function for the ith (j^{th}) participant (collision) source where,

$$F_{i,j}(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-x_{i,j})^2+(y-y_{i,j})^2}{2\sigma^2}}. \quad (3)$$

The smearing radius around each participant source (σ) is considered as 0.4 fm [36,55] and the overall normalization constant K in Eqn. (2) is adjusted to produce charged particle multiplicity ($dn_{\rm ch}/d\eta$) for most central O+O collisions ∼180 [18].

The initial state for each random event is then subsequently evolved with a longitudinally boost invariant (2+1) ideal relativistic hydrodynamic framework to obtain the space-time evolutions at the mid rapidity. The MUSIC hydrodynamical framework has been used for this study, which has been used extensively to reproduce the charged particle spectra and anisotropic flow in relativistic nuclear collisions [56–58].

The initial formation time τ_0 of the plasma is taken as 0.14 fm/c and a lattice based EOS from Ref [59] is used. The τ_0 is taken as the same for Pb+Pb collisions at 2.76A TeV [25].

A precise determination of the thermalization time τ_0 is only possible through model calculation [60,61] and a detailed comparison with experimental data which are not yet available for O+O collisions at present. It is also reasonable to expect a larger value of τ_0 for smaller systems such as O+O and earlier studies have shown that both the production and anisotropic flow of thermal photons are sensitive to the choice of τ_0 [22]. However, the qualitative nature of photon anisotropic flow remains unchanged with varying τ_0 and the ratio of photon flow coefficients is found to be only weakly affected by larger values of τ_0 [33].

It has been shown that most central p+Pb collisions at 2.76A TeV at the LHC result in significant thermal radiation and photon anisotropic flow [62]. This suggests that O+O collisions at 7A TeV may develop sufficient collectivity to produce appreciable thermal radiation, although the thermalization time and pre-equilibrium contribution in such small systems still need to be addressed properly.

In this initial study we thus retain the same value of τ_0 used in earlier LHC analyses and a systematic investigation of the sensitivity of our results to τ_0 is postponed to future work.

A constant freeze-out temperature of 137 MeV is considered for both the clustered and unclustered collisions and impact parameter range 0-−1.37 and 2.6-−3.0 fm respectively for 0–5% and 30–40% collision centrality bins [50]. It is to be noted that in experimental analyses, fixing the impact parameter for a given centrality class is not a clean cut procedure, as centrality bins are determined a posteriori using the charged particle multiplicity distribution. In the present exploratory study, we adopt a fixed impact parameter approach to represent different centrality classes in order to reduce the computational cost.

The production of thermal photons is estimated by integrating the emission rates ($R = E\frac{dN}{d^3pd^4x}$) from QGP and hot hadronic matter over the space-time history:

$$E\frac{dN}{d^3p} = \int R\big(E^*(x), T(x)\big) d^4x. \quad (4)$$

$T(x)$ in the above equation is the local temperature. $E^*(x) = p^\mu u_\mu(x)$, where p^μ represents the 4-momentum of photon and u_μ is the local 4-velocity of the flow field. We have used the NLO QGP rates from Refs. [64,65] and parametrized rates of photon production from Ref. [66] to estimate the production and anisotropic flow of thermal photons at 7A TeV at the LHC.

The JETPHOX framework has been employed to estimate NLO pQCD photon production at $\sqrt{s_{\rm NN}} = 7$ TeV [67]. The calculations use the CT14 parton distribution function [68] together with EPS09 [69] parametrization of nuclear shadowing and the BFGII fragmentation function [70]. The renormalization, factorization, and fragmentation scales are all set to p_T. An isolation cone of radius $R = 0.4$ and a transverse momentum cutoff of $p_T^{\rm cut} \approx 0.1 p_T$ are applied to evaluate the total prompt photon production [71]. Results for the 0—5% and 30—40% centrality classes are obtained by scaling with the corresponding numbers of binary collisions taken to be approximately 53 and 24 respectively.

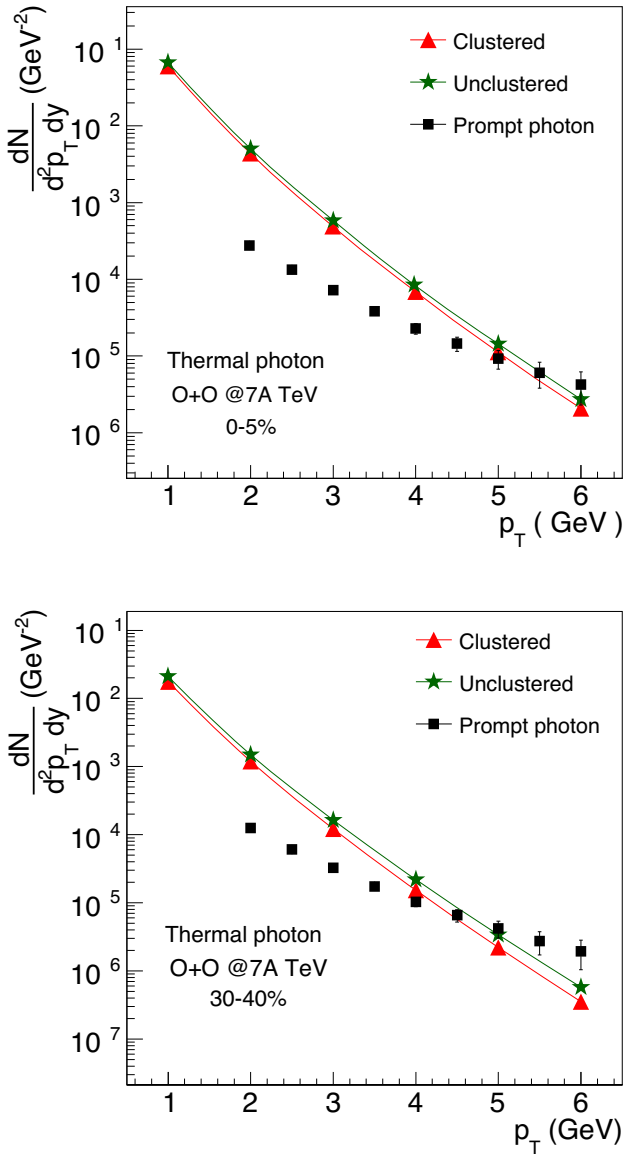


Fig. 3 Thermal photon spectra from 7A TeV O+O collisions at LHC for 0–5% [upper panel] and 30–40% [lower panel] centrality bins. The prompt photon spectra for the same centrality bins are shown for a comparison

The differential anisotropic flow coefficients $v_n(p_T)$ ($n = 2$ and 3) for each event is obtained as:

$$v_n(p_T) = \frac{\int_0^{2\pi} d\phi \cos[n(\phi - \psi_n)] \frac{dN}{p_T dp_T dy d\phi}}{\int_0^{2\pi} d\phi \frac{dN}{p_T dp_T dy d\phi}}. \quad (5)$$

Where, ψ_n is the participant plane angle and ϕ is the azimuthal angle of particle momentum. The participant plane angle is estimated using the relation,

$$\psi_n = \frac{1}{n} \arctan \frac{\int dx dy\, r^2 \sin(n\Phi)\, \epsilon(x, y, \tau_0)}{\int dx dy\, r^2 \cos(n\Phi)\, \epsilon(x, y, \tau_0)} + \pi/n, \quad (6)$$

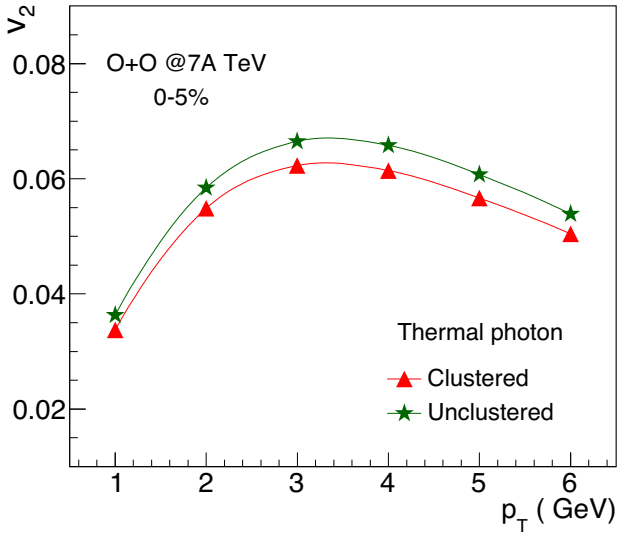


Fig. 4 Elliptic flow of thermal photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 0–5% centrality bin

In the present work, we estimate photon anisotropic flow using the participant plane angle. While the scalar product method would provide a more direct connection to experimental measurements, it is computationally much more demanding. It has been shown that in the presence of hydrodynamically evolving density fluctuations, the participant plane provides a good approximation to the event plane for hadronic elliptic flow [63]. It is to be noted that the photon anisotropic flow results from scalar product methods are found to be similar to those obtained from the participant plane method [36]. Additionally, the inclusion of viscosity is essential for a more quantitative comparison with experimental data. However, the qualitative features of the results due to the presence of α clustered structures are expected to remain unchanged under viscous hydrodynamic expansion.

3 Results and discussion

The initial spatial eccentricities ϵ_2 and ϵ_3 for α clustered and unclustered O+O collisions at 7A TeV at the LHC as a function of wounded nucleons ($N_{\rm wn}$) are shown in Figs. 1 and 2 respectively. The spatial eccentricities are obtained by averaging over 10^6 random events for each configuration. The ϵ_2 is found to be small but nonzero even for the most central collisions for both cases arising from initial state fluctuations. In addition, the ϵ_2 in the most central collisions is smaller for the clustered case than the unclustered one, reflecting the reduced role of fluctuations in the presence of a correlated clustered structure. Whereas, in mid-central collisions, the ϵ_2 is slightly larger for the clustered collisions than for the unclustered collisions.

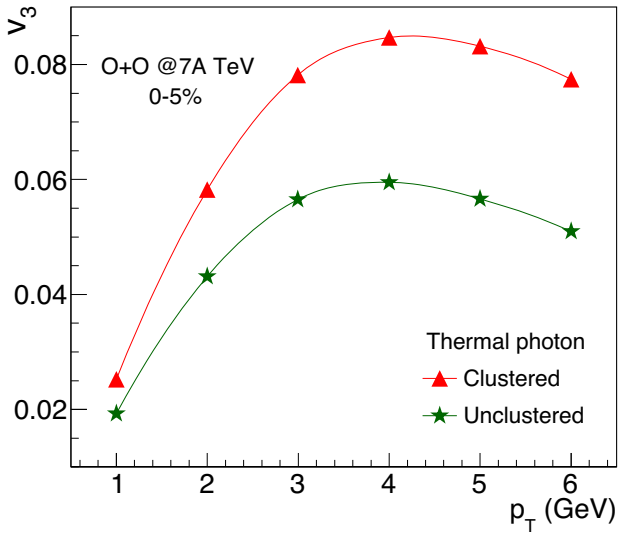


Fig. 5 Triangular flow of thermal photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 0–5% centrality bin

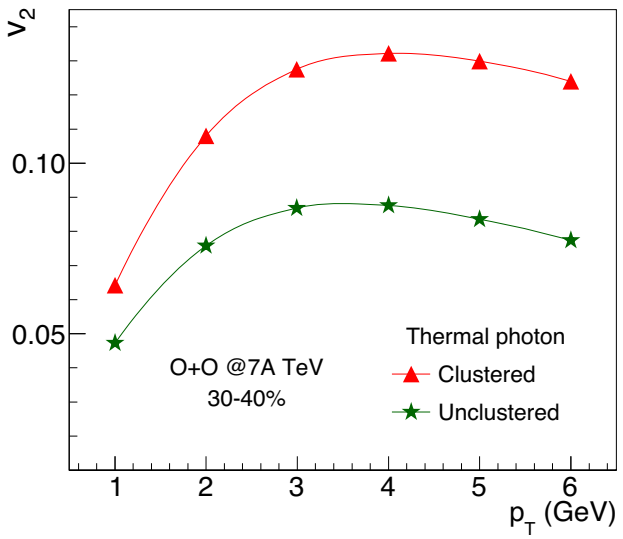


Fig. 6 Elliptic flow of thermal photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 30–40% centrality bin

The value of ϵ_3 is found to be significantly larger for the clustered configuration compared to the unclustered one for central collisions [see Fig. 2]. In contrast, for mid-central and peripheral collisions, the initial triangular eccentricity shows no noticeable dependence on the clustered structure.

The ϵ_3 exhibits weaker dependence to collision centrality compared to ϵ_2 except for very peripheral events.

3.1 Photon spectra

The thermal photon spectra from clustered and unclustered O+O collisions at 7A TeV are shown in Fig. 3. The thermal spectra are estimated for two centrality bins, 0—5% and 30—40%. The prompt photon spectra for both centrality bins are shown for a comparison.

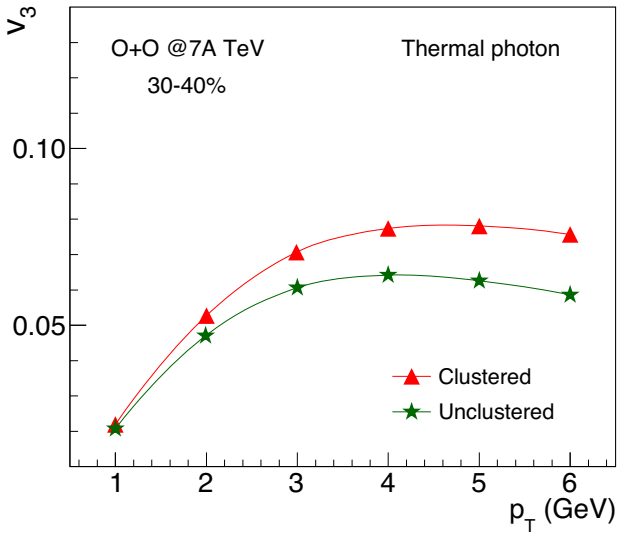


Fig. 7 Triangular flow of thermal photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 30–40% centrality bin

The presence of clustered structure in oxygen nuclei is found to have only a marginal effect on photon production in the low p_T (below 3 GeV) region for both central and peripheral collisions. The effect of initial state fluctuations is expected to be more pronounced for larger p_T region, which might result in a small difference in the spectra from clustered and unclustered cases for $p_T > 3$ GeV.

The fluctuations may be more significant in the unclustered case due to the random distribution of nucleons in the small volume, which may lead to an enhanced photon production relative to clustered collisions.

3.2 Anisotropic flow of thermal photons

The anisotropic flow of thermal photons from 0–5% central O+O collisions at 7A TeV at the LHC is shown in Figs. 4 and 5 for both clustered and unclustered collisions. The anisotropic flow results are obtained by taking an average over 400 random events. The elliptic flow parameter as a function of p_T shows similar nature to the $v_2(p_T)$ of photons from heavy ion collisions at RHIC and LHC energies [31].

For the 0—5% centrality bin, the elliptic flow is found to be significantly larger in the unclustered case due to initial state fluctuations. In contrast, the presence of a clustered configuration suppresses these fluctuations due to correlations among the clustered sites. Consequently, both the initial eccentricity ϵ_2 and the resulting elliptic flow $v_2(p_T)$ are found to be smaller in the clustered case compared to the unclustered scenario.

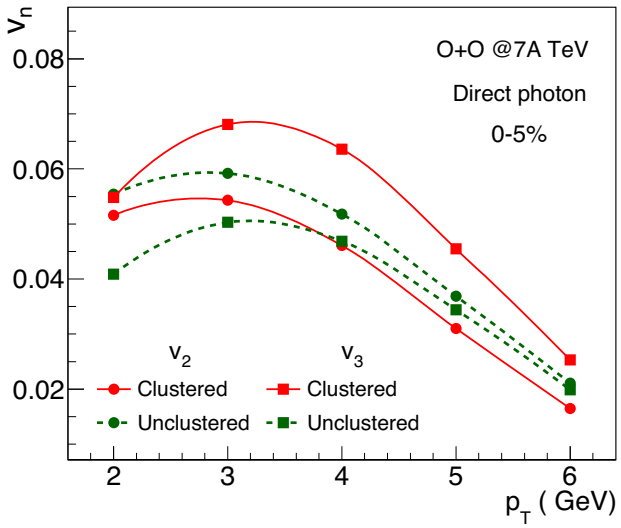


Fig. 8 Elliptic and triangular flow of direct (thermal + prompt) photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 0–5% centrality bin

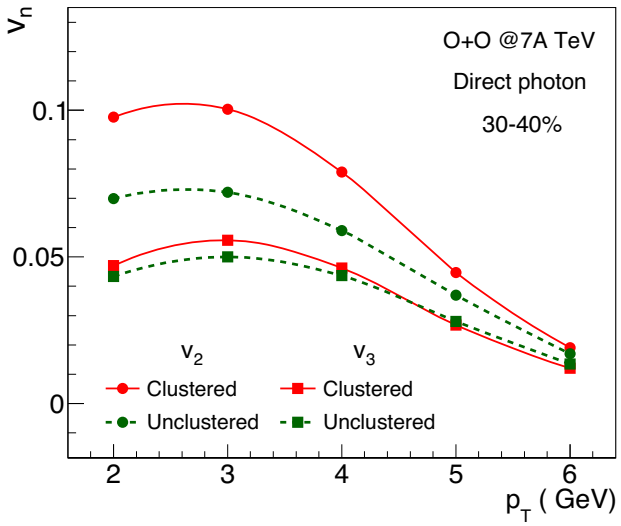


Fig. 9 Elliptic and triangular flow of direct (thermal + prompt) photons from clustered and unclustered O+O collisions at 7A TeV at LHC for 30–40% centrality bin

The elliptic and triangular flow exhibit distinct qualitative behavior for the clustered and unclustered configurations for 0–5% centrality bin. The p_T dependent triangular flow parameter is significantly larger than the elliptic flow coefficient in the clustered configuration. However, for the unclustered (or uniform) case the elliptic flow is considerably larger than the triangular flow $v_3(p_T)$. These anisotropic flow patterns can be understood from the eccentricity plots presented earlier, the larger ϵ_3 in the clustered case and the relatively larger ϵ_2 in the unclustered case primarily decide the observed anisotropic flow results.

For the peripheral collisions of 30–40% centrality bin, the elliptic as well as triangular flow are both found to be larger for the clustered configuration than unclustered collisions (Figs. 6 and 7).

Additionally, the $v_4(p_T)$ coefficient of thermal photons is found to be negligible even though the initial ϵ_4 is significantly large for both the collision centralities [not shown here]. Thus, the tetrahedral α clustered structure in the oxygen nucleus is found to affect the triangular flow parameter maximum.

The direct photon anisotropic flow ($v_n^{\rm dir}$) can be estimated by including the prompt contribution to the thermal one using the relation;

$$v_n^{\rm dir}(p_T) = \frac{v_n^{\rm th}(p_T) \times dN^{\rm th}(p_T)}{dN^{\rm th}(p_T) + dN^{\rm pr}(p_T)} \tag{7}$$

Where, $v_n^{\rm th}$ and $dN^{\rm th}$ are the anisotropic flow and yield of thermal photons respectively at a certain p_T and $dN^{\rm pr}$ is the yield of prompt photons at that p_T. The prompt photons do not contribute directly to anisotropic flow however, they add extra weight factor in the denominator of direct photon v_n calculation. The direct photons v_n results are shown in Figs. 8 and 9 respectively. The inclusion of prompt photons brings the overall v_n down towards lower p_T values. However, the qualitative difference between the anisotropic flow parameters due to initial state clustered structure compared to the unclustered one remain unchanged.

3.3 Ratio of v_2 and v_3

It has been shown that the ratio of elliptic and triangular flow parameters of direct photons is a valuable observable in the study of initial state and thermal radiation produced in relativistic heavy ion collisions. The ratio as a function of p_T can significantly suppress uncertainties arising from non-thermal contributions in the analysis of photon anisotropic flow [33].

The ratio of the elliptic and triangular flow parameters (v_2/v_3) as a function of p_T is shown in Figs. 10 and 11 for O+O collisions at 7A TeV. As expected from the individual v_n results, the ratio from clustered collisions is smaller than 1 for $p_T > 1$ GeV. Whereas, the same ratio is larger than 1 for the unclustered case in the entire p_T region. The results from peripheral collisions show that the ratio is always greater than 1 as $v_2(p_T)$ is always larger than $v_3(p_T)$ for both configurations. Additionally it has also been shown in [33] that the ratio of photon anisotropic flow parameters is significantly less sensitive to the initial formation time of the plasma compared to the spectra and the individual anisotropic flow parameters of thermal photons.

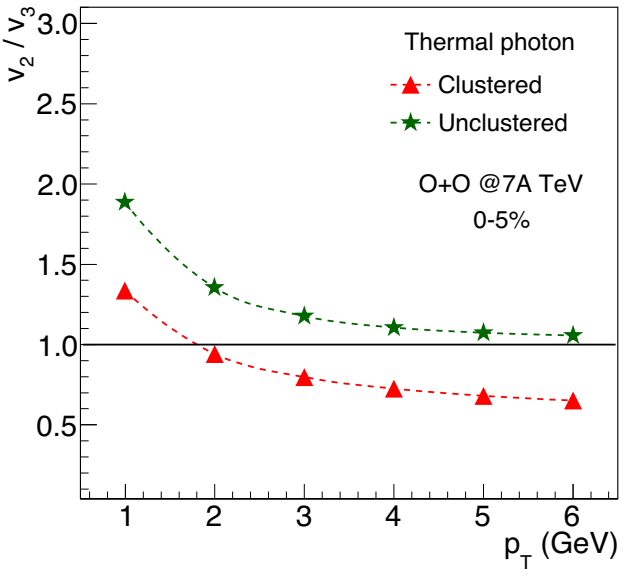


Fig. 10 Ratio of elliptic and triangular flow of thermal photons in clustered and unclustered O+O collisions at 7A TeV at LHC for 0–5% centrality bin

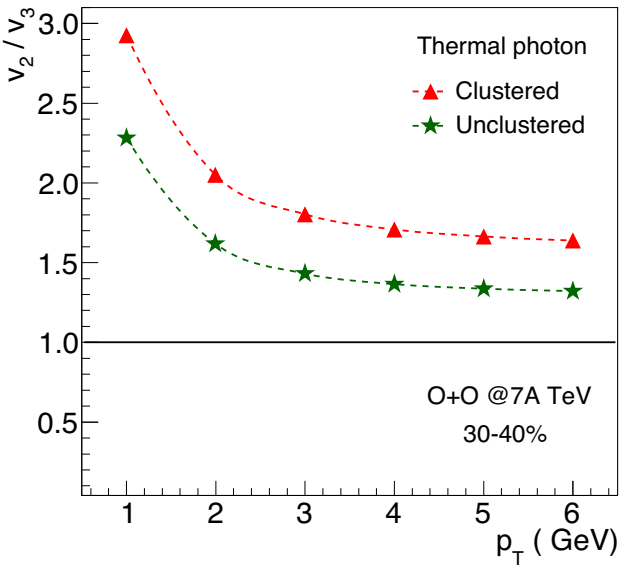


Fig. 11 Ratio of elliptic and triangular flow of thermal photons in clustered and unclustered O+O collisions at 7A TeV at LHC for 30–40% centrality bin

4 Summary and conclusions

The GLISSANDO initial conditions combined with the MUSIC hydrodynamical model have been used to study the initial state, photon production, and anisotropic flow parameters of thermal photons in 7A TeV O+O collisions at the LHC. The O+O collisions, being a small system, exhibit sizable initial state eccentricities from event-by-event fluctuations even in the most central events, leading to large ϵ_n $(n = 2, 3)$ for both clustered and unclustered cases. These fluctuations are expected to be more pronounced in the unclustered configuration.

However, the presence of a tetrahedral α clustered structure in oxygen nuclei is found to significantly affect the initial triangular eccentricity in the most central collisions compared to the unclustered O+O collisions.

The photon spectra from clustered and unclustered configurations are found to be close to each other for both central and peripheral collisions. Only at $p_T > 3$ GeV, the photon production from unclustered O+O collisions is found to be slightly larger than the clustered one.

The triangular flow of thermal photons in clustered O+O collisions is found to be significantly larger than $v_3(p_T)$ in the unclustered configuration for the most central collisions. In contrast, the elliptic flow in the unclustered case is slightly higher than that in clustered collisions. For peripheral collisions however, both anisotropic flow coefficients are larger in the α clustered configuration compared to the unclustered case. The elliptic flow shows a strong centrality dependence increasing substantially from central to peripheral collisions whereas, the triangular flow exhibits little sensitivity to collision centrality.

The ratio of photon v_2 and v_3 is shown to be a robust observable in which non-thermal contributions are minimized and which is relatively less sensitive to uncertainties in the model calculations compared to the thermal photon v_n and p_T spectra. Thus, the ratio of photon v_n as a function of p_T for different centrality bins is expected to exhibit qualitatively different behavior due to the presence of the clustered structure which could be verified experimentally.

In conclusion, the production and anisotropic flow of direct photons in O+O collisions at the LHC can serve as sensitive probes of the underlying nuclear structure. Our results indicate that anisotropic flow observables are significantly influenced by the presence of cluster structures in the oxygen nuclei suggesting that direct photon measurements can be a powerful tool for exploring sub-nuclear structure in light nuclei.

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Code Availability Statement This code/software will be made available on reasonable request. [Authors' comment: The codes for MUSIC and GLISSANDO-3 used in this work are publicly available].

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